

# ST-911 Schema Delta Analysis

## V5 Telemetry CSV → Current Parser Gap Report

**Date:** 2026-05-14  
**Source CSVs:** `s2s_telemetry_20260511_173825.csv`, `s2s_telemetry_20260511_204358.csv`  
**Both CSVs have identical 70-column headers.**

### 1. Column Renames (BREAKING — parser will silently drop these columns)

| V4 COL_MAP Key       | V5 CSV Header            | Internal Name        | Impact                                                                                                                                   |
|----------------------|--------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <code>alt_msl</code> | <code>alt_msl_ft</code>  | <code>alt_msl</code> | <b>CRITICAL</b> — used by phase-detector, arc1-grader, go-around. Currently maps to nothing → all altitude checks get zeros.             |
| <code>alt_agl</code> | <code>alt_agl_ft</code>  | <code>alt_agl</code> | <b>CRITICAL</b> — used everywhere for AGL thresholds. Same zero-fill problem.                                                            |
| <code>hdg_deg</code> | <code>hdg_mag_deg</code> | <code>hdg</code>     | <b>CRITICAL</b> — all heading-based pattern detection (down-wind, base, final, crosswind) uses <code>col(df, 'hdg')</code> . Gets zeros. |

### Fix Required

Add V5 aliases to `COL_MAP` :

```
alt_msl_ft: 'alt_msl',    // V5 rename
alt_agl_ft: 'alt_agl',    // V5 rename
hdg_mag_deg: 'hdg',       // V5 rename (magnetic heading)
```

## 2. New V5 Columns NOT in COL\_MAP (need to be added)

| V5 CSV Header    | Suggested Internal Name | Used By                                               | Priority |
|------------------|-------------------------|-------------------------------------------------------|----------|
| hdg_true_deg     | hdg_true                | Not currently used, but useful for future nav scoring | LOW      |
| beta_deg         | beta                    | Sideslip angle — potential slip/skid scoring          | LOW      |
| rep_preheat      | rep_preheat             | REP engine state for preflight checks                 | MEDIUM   |
| rep_plug_fouling | rep_plug_fouling        | REP engine state — runup grading                      | MEDIUM   |
| rep_primer       | rep_primer              | REP engine start procedure                            | MEDIUM   |
| rep_magneto      | rep_magneto             | REP magneto check — runup grading                     | MEDIUM   |
| recorder_version | recorder_version        | Schema version detection (per LOGIC_DELTA)            | HIGH     |

### Fix Required

Add these to COL\_MAP :

```

hdg_true_deg: 'hdg_true',
beta_deg: 'beta',
rep_preheat: 'rep_preheat',
rep_plug_fouling: 'rep_plug_fouling',
rep_primer: 'rep_primer',
rep_magneto: 'rep_magneto',
recorder_version: 'recorder_version',

```

## 3. lib/telemetry-parser.ts (ST-505 legacy parser) — Additional Gaps

This older parser uses EXPECTED\_COLUMNS with **different naming conventions** that don't match V5:

| EXPECTED_COLUMNS Entry | V5 CSV Header | Status                  |
|------------------------|---------------|-------------------------|
| alt_msl                | alt_msl_ft    | ✗ MISSING — needs alias |
| alt_agl                | alt_agl_ft    | ✗ MISSING — needs alias |
| hdg_deg                | hdg_mag_deg   | ✗ MISSING — needs alias |
| beacon_on              | light_beacon  | ✗ Name mismatch         |
| avionics_on            | rep_avionics  | ✗ Name mismatch         |

Many V5 columns (REP engine params, head\_roll, wind, lights, etc.) are completely absent from EXPECTED\_COLUMNS .

### Fix Required

Update EXPECTED\_COLUMNS to use V5 names OR add a column-alias normalization step before lookup.

## 4. types/telemetry.ts — TelemetryRow Interface Gaps

The TelemetryRow interface is missing many V5 fields and uses stale names:

| Interface Field                                                                                                                                                                                                                       | V5 CSV Header | Status             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|
| alt_msl                                                                                                                                                                                                                               | alt_msl_ft    | ✗ Stale name       |
| alt_agl                                                                                                                                                                                                                               | alt_agl_ft    | ✗ Stale name       |
| hdg_deg                                                                                                                                                                                                                               | hdg_mag_deg   | ✗ Stale name       |
| beacon_on                                                                                                                                                                                                                             | light_beacon  | ✗ Stale name       |
| avionics_on                                                                                                                                                                                                                           | rep_avionics  | ✗ Stale name       |
| Missing: alpha_deg ,<br>beta_deg , carb_heat ,<br>flap_handle_dep ,<br>fuel_total_kg , g_side ,<br>gear_deflect_mtr , generat-<br>or_on , head_roll , towbar ,<br>wind_spd_kt , wind_dir_deg ,<br>rep_* fields, record-<br>er_version | —             | ✗ Not in interface |

### Fix Required

Expand TelemetryRow to include all 70 V5 columns with correct naming.

## 5. Downstream Consumer Impact

All downstream files reference internal names via `col(df, 'name')`. As long as `COL_MAP` maps V5 headers → existing internal names, **no downstream changes are needed** for the 3 critical renames:

| File                                   | References                                                     | Impact After COL_MAP Fix |
|----------------------------------------|----------------------------------------------------------------|--------------------------|
| <code>phase-detector.ts</code>         | <code>alt_agl</code> , <code>alt_msl</code> , <code>hdg</code> | ✓ No change needed       |
| <code>arc1-grader.ts</code>            | <code>alt_agl</code> , <code>alt_msl</code> , <code>hdg</code> | ✓ No change needed       |
| <code>landing-scorer.ts</code>         | <code>gear_deflect</code>                                      | ✓ Already mapped         |
| <code>go-around.ts</code>              | <code>alt_agl</code> , <code>hdg</code>                        | ✓ No change needed       |
| <code>gtgl-sequence-detector.ts</code> | <code>hdg</code> (via metadata)                                | ✓ No change needed       |

## 6. Summary of Required Changes (ST-911 Scope)

### Priority 1 — CRITICAL (parser breaks without these)

1. `lib/s2s-engine/telemetry-parser.ts` **COL\_MAP**: Add 3 V5 renames ( `alt_msl_ft`, `alt_agl_ft`, `hdg_mag_deg` )
2. `lib/s2s-engine/telemetry-parser.ts` **COL\_MAP**: Add 7 new V5 columns ( `hdg_true_deg`, `beta_deg`, `rep_preheat`, `rep_plug_fouling`, `rep_primer`, `rep_magneto`, `recorder_version` )

### Priority 2 — TYPE SAFETY

1. `types/telemetry.ts` : Update `TelemetryRow` interface to match V5 schema
2. `lib/telemetry-parser.ts` : Update `EXPECTED_COLUMNS` to include V5 column names

### Priority 3 — TESTING (per ST-911 ticket)

1. **Unit tests**: Verify V5 CSV headers parse without errors
2. **Integration smoke test**: Full `evaluateFlight()` pass with sample CSVs
3. `PROJECT_CHANGELOG.md` : Append v1.0.1-beta entry

## 7. Backward Compatibility Note

The V5 `COL_MAP` additions are **additive** — existing V4 keys ( `alt_msl`, `alt_agl`, `hdg_deg` ) should be **kept** in `COL_MAP` so older CSV files still parse correctly. The parser maps whichever header it finds first.